

Podophyllum peltatum (Berberidaceae) American Mandrake

Description Perennial plant growing to 1 6 in (40 cm). Has a forked stem, two deeply lobed umbrella-like leaves, white flowers, and small yellow fruit.

Habitat & Cultivation American mandrake is native to northeastern North America. It is commonly found in damp woodland and pastureland. The rhizome is unearthed in autumn.

Part Used Rhizome.

Constituents The rhizome of American mandrake contains lignans (especially podophyllotoxin), flavonoids, resin, and gums.

The lignans are responsible for the rhizome's purgative action.

History & Folklore

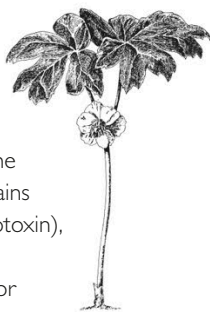
American mandrake was much used as a purgative, emetic, and worm-expelling herb by Native Americans. In the U.S. in the 19th century, both herbal and conventional medical practitioners regarded the plant as the safest and most readily available purge.

Medicinal Actions & Uses Despite 19th-century beliefs in its safety, American mandrake is no longer taken internally on account of its cytotoxic (cell-killing) action. However, applied externally as a poultice, lotion, or ointment, the root can be an effective treatment for all kinds of warts.

Research The lignans in American mandrake—podophyllotoxin in particular—act against tumors and have been extensively researched for their anticancer potential. Semisynthetic derivatives of podophyllotoxin appear to be the most promising, having minimal toxicity.

Related Species The Himalayan *P. hexandrum* may have similar actions.

⚠️ Cautions Do not take internally. The plant is subject to legal restrictions in most countries.



American mandrake

Pogostemon cablin syn. *P. patchouli* (Lamiaceae)

Patchouli

Description Hairy, aromatic perennial growing to 3 ft (1 m). Has square stems, oval leaves, and spikes bearing whorls of white to light-purple flowers.

Habitat & Cultivation Native to Malaysia and the Philippines, patchouli is now cultivated in tropical and subtropical regions around the world. The shoots and leaves are picked 2 or 3 times a year.

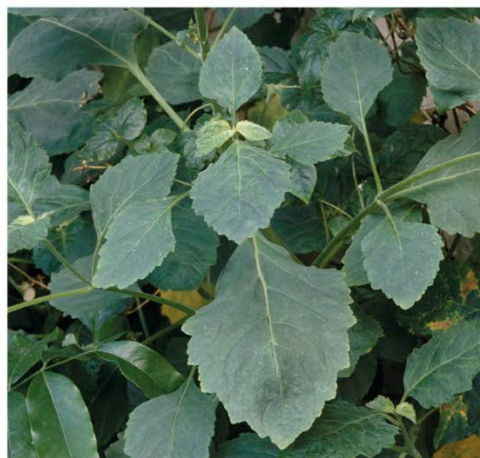
Parts Used Young leaves and shoots, essential oil.

Constituents Patchouli contains a volatile oil comprising mainly the sesquiterpenes patchoulol (35%) and bulnesene.

History & Folklore Patchouli has been used extensively in Asian medicine, featuring in the Chinese, Indian, and Arabic traditions. Its most common use has been as an aphrodisiac. The essential oil is widely employed in India as a fragrance and as an insect repellent.

Medicinal Actions & Uses Patchouli is used in herbal medicine in Asia as an aphrodisiac, antidepressant, and antiseptic. It is also employed for headaches and fever. Patchouli essential oil is used in aromatherapy to treat skin complaints. It is thought to have a regenerative effect on skin tone and to help clear conditions such as eczema and acne. The oil may also be used for varicose veins and hemorrhoids.

Caution Do not take essential oil internally.



Patchouli is the source of an essential oil.

Polygala senega (Polygalaceae) Seneca Snakeroot

Description Perennial growing to about 1 6 in (40 cm). Has narrow lance-shaped leaves with toothed edges, and spikes of whitish-pink flowers.

Habitat & Cultivation Seneca snakeroot is native to North America, and found in dry stony open ground and woodland. It is cultivated in western Canada. The root is unearthed in autumn.

Part Used Root.

Constituents Seneca snakeroot contains phenolic acids, methyl salicylate, triterpenoid saponins (including sengins), polygalitol, and plant sterols. The triterpenoid saponins promote the clearing of phlegm from the bronchial tubes.

History & Folklore This plant's name refers to the Seneca tribe of North America, who employed the root as a remedy for snake bite. Seneca snakeroot was highly valued as a medicine by both Native Americans and European settlers. In 1768, Dr. Alexander Garden of Charleston wrote that "The

Seneca is the most powerful and efficacious antiphlogistic [fever- and inflammation-reducing substance] attenuant among the Galenical medicines."

Medicinal Actions & Uses In North American and European herbal medicine, Seneca snakeroot is used as an expectorant to treat bronchial asthma, chronic bronchitis, and whooping cough. The root has a stimulant action on the bronchial mucous membranes, promoting the coughing up of phlegm from the chest. Seneca snakeroot is also thought to encourage sweating and to stimulate the secretion of saliva.

Related Species *Yuan zhi* (*P. tenuifolia*), native to China and Japan, has similar constituents. *Yuan zhi* is taken to treat congestion in the chest and to "calm the spirit and quieten the heart." See also milkwort (*P. vulgaris*, following entry).

Cautions Do not take if pregnant. Causes diarrhea and vomiting in excessive doses.

Polygala vulgaris (Polygalaceae) Milkwort

Description Short perennial with pointed lance-shaped leaves and spikes of small blue, mauve, or white flowers.

Habitat & Cultivation Milkwort is common in grassy and moorland areas in much of western and northern Europe. It is gathered from the wild when the plant is in flower in summer.

Parts Used Aerial parts, root.

Constituents Milkwort contains triterpenoid saponins, a volatile oil, gaultherin, and mucilage.

History & Folklore Milkwort has been most often used to treat chest problems such as pleurisy and dry coughs. In larger doses, the plant acts as an emetic. In his *Irish Herbal* (1735), K'Eogh states that "it has a hot dry nature, and it encourages the production of milk in nursing mothers."

Medicinal Actions & Uses While milkwort is infrequently used in European herbal medicine today, it—like Seneca snakeroot (*P. senega*, see preceding entry)—is a valuable herb for the treatment of respiratory troubles such as chronic bronchitis, bronchial asthma, and convulsive coughs, including whooping cough. Milkwort is also thought to have sweat-inducing and diuretic properties.

Polygonatum multiflorum (Liliaceae)

Solomon's Seal

Description Perennial growing to about 20 in (50 cm). Has arching stems, alternate elliptical leaves, delicate greenish-white, bell-shaped flowers, and blue-black fruit.